```
import pandas as pd
from google.colab import files
```

```
uploaded = files.upload()
```

Choose Files marketing_campaign.xlsx
marketing_campaign.xlsx(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) - 315875 bytes, last modified: 6/11/2026 - 100% done
Saving marketing_campaign.xlsx to marketing_campaign.xlsx

```
df = pd.read_excel('marketing_campaign.xlsx')
df.head()
```

	ID	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Dt_Customer	Recency	MntWines	...	NumWebVisits
0	5524	1957	Graduation	Single	58138.0	0	0	2012-09-04	58	635	...	
1	2174	1954	Graduation	Single	46344.0	1	1	2014-03-08	38	11	...	
2	4141	1965	Graduation	Together	71613.0	0	0	2013-08-21	26	426	...	
3	6182	1984	Graduation	Together	26646.0	1	0	2014-02-10	26	11	...	
4	5324	1981	PhD	Married	58293.0	1	0	2014-01-19	94	173	...	

5 rows × 29 columns

```
df.shape
```

(2240, 29)

```
df.isnull().sum()
```

	0
ID	0
Year_Birth	0
Education	0
Marital_Status	0
Income	24
Kidhome	0
Teenhome	0
Dt_Customer	0
Recency	0
MntWines	0
MntFruits	0
MntMeatProducts	0
MntFishProducts	0
MntSweetProducts	0
MntGoldProds	0
NumDealsPurchases	0
NumWebPurchases	0
NumCatalogPurchases	0
NumStorePurchases	0
NumWebVisitsMonth	0
AcceptedCmp3	0
AcceptedCmp4	0
AcceptedCmp5	0
AcceptedCmp1	0
AcceptedCmp2	0
Complain	0
Z_CostContact	0
Z_Revenue	0
Response	0

dtype: int64

```
df['Income'].fillna(df['Income'].median(), inplace=True)
```

```
/tmp/ipykernel_2481/2154417476.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through c
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]


  df['Income'].fillna(df['Income'].median(), inplace=True)
```

```
df.isnull().sum()
```

	0
ID	0
Year_Birth	0
Education	0
Marital_Status	0
Income	0
Kidhome	0
Teenhome	0
Dt_Customer	0
Recency	0
MntWines	0
MntFruits	0
MntMeatProducts	0
MntFishProducts	0
MntSweetProducts	0
MntGoldProds	0
NumDealsPurchases	0
NumWebPurchases	0
NumCatalogPurchases	0
NumStorePurchases	0
NumWebVisitsMonth	0
AcceptedCmp3	0
AcceptedCmp4	0
AcceptedCmp5	0
AcceptedCmp1	0
AcceptedCmp2	0
Complain	0
Z_CostContact	0
Z_Revenue	0
Response	0

dtype: int64

```
df['Total_Spending'] = (
      df['MntWines'] +
          df['MntFruits'] +
              df['MntMeatProducts'] +
                  df['MntFishProducts'] +
                      df['MntSweetProducts'] +
                          df['MntGoldProds']
                          )
```

```
df[['Total_Spending']].head()
```

	Total_Spending
0	1617
1	27
2	776
3	53
4	422

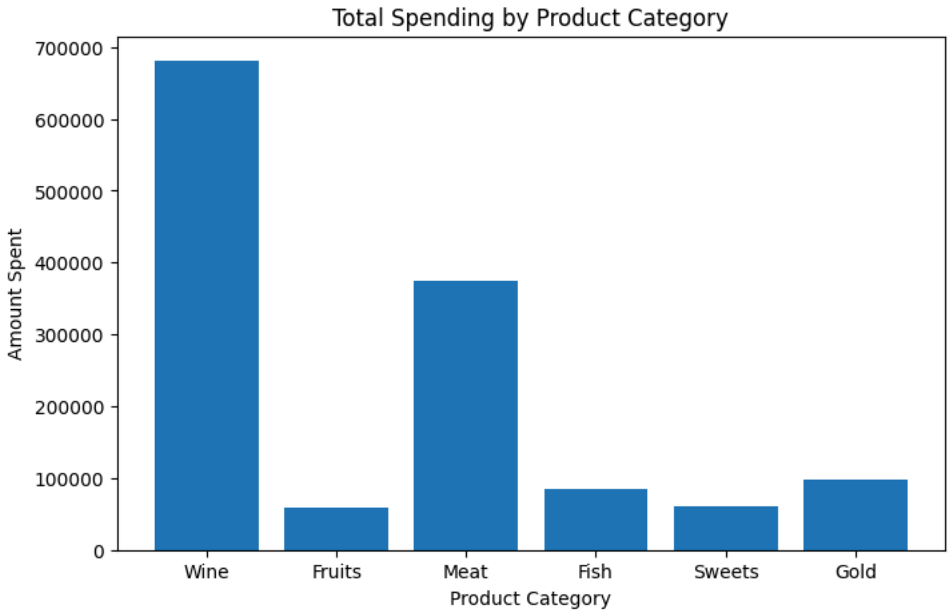
```
print("Total Customers:", len(df))
print("Average Income:", round(df['Income'].mean(), 2))
print("Average Spending:", round(df['Total_Spending'].mean(), 2))
```

```
Total Customers: 2240
Average Income: 52237.98
Average Spending: 605.8
```

```
import matplotlib.pyplot as plt

products = ['Wine', 'Fruits', 'Meat', 'Fish', 'Sweets', 'Gold']
values = [
    df['MntWines'].sum(),
        df['MntFruits'].sum(),
            df['MntMeatProducts'].sum(),
                df['MntFishProducts'].sum(),
                    df['MntSweetProducts'].sum(),
                        df['MntGoldProds'].sum()
                        ]

plt.figure(figsize=(8,5))
plt.bar(products, values)
plt.title('Total Spending by Product Category')
plt.xlabel('Product Category')
plt.ylabel('Amount Spent')
plt.show()
```

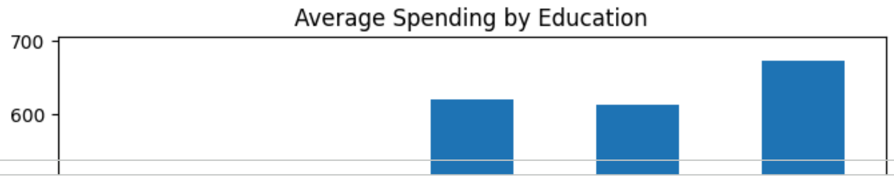



```
education_spending = df.groupby('Education')['Total_Spending'].mean()

import matplotlib.pyplot as plt

education_spending.plot(kind='bar', figsize=(8,5))

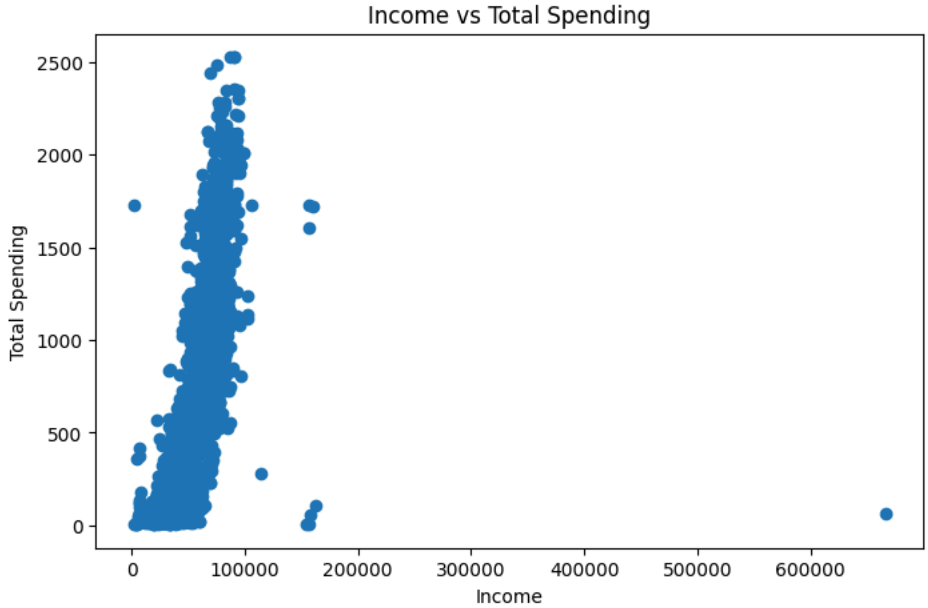
plt.title('Average Spending by Education')
plt.xlabel('Education Level')
plt.ylabel('Average Spending')
plt.show()
```




```
plt.figure(figsize=(8,5))
plt.scatter(df['Income'], df['Total_Spending'])

plt.title('Income vs Total Spending')
plt.xlabel('Income')
plt.ylabel('Total Spending')

plt.show()
```




```
print("Total Customers:", len(df))
print("Average Income:", round(df['Income'].mean(), 2))
print("Average Spending:", round(df['Total_Spending'].mean(), 2))
```

```
Total Customers: 2240
Average Income: 52237.98
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